

Max-Planck-Institut für Hirnforschung, Frankfurt/Main-Niederrad, BRD,
and Université de Montréal, Département des Sciences biologiques, Montréal, Canada

Comparison of Brain Structure Volumes in Insectivora and Primates VII. Amygdaloid Components

Heinz STEPHAN, Heiko D. FRAHM and Georg BARON

With 14 Figures and 3 Tables

(Received September 4, 1986)

Abstract and Summary: Volumes of the amygdaloid complex (AMY) and some subdivisions were measured in 2 species of Macroscelidea, 39 species of Insectivora, 3 species of Scandentia, 18 species of prosimians, 26 species of simians and man. Changes in the relative size from Insectivora through man (expressed by size indices) showed a definite tendency to increase in the cortico-basolateral subdivision (LAM) and more or less constant relations in the centromedial subdivision (MAM), except for the nucleus tractus olfactorii lateralis (NTO), which becomes distinctly reduced. The reduction of NTO is even stronger than that of the main olfactory bulb (BOL), which in simians and man is small but distinct, whereas NTO is hardly recognizable in most of these forms. In the LAM group the size increase of the large-celled part of the basal nucleus (MCB) is less than that of the small-celled components (including the lateral nucleus). The differences between LAM and MAM groups are discussed with regard to the predominant fiber connections, which in MAM are stronger with conservative brains parts (brainstem), and in LAM with progressive brain parts (e.g. neocortex).

Key words: Morphometry — Amygdala — Olfactory system — Limbic system — Insectivora — Primates

Introduction

Many structures in the mammalian amygdaloid complex have been described and subsequently identified by a variety of names such as nuclei, accessory nuclei, subnuclei, complexes and areas. Most of these structures cannot be accurately delineated and homologized in all species of a large sample and, therefore, cannot be compared quantitatively in a comprehensive study. However, in all species from Insectivora through Primates, it has been possible to distinguish between two amygdaloid components, one of which (cortico-basolateral group or simply lateral amygdala, LAM) combines the lateral (L), basal (B, including a magnocellular part, MCB) and cortical (CO) nuclei, and the other (centromedial group or simply medial amygdala, MAM), combines the central (CE) and medial (M) nuclei and the anterior amygdaloid area (AA) enclosing the nucleus of the lateral olfactory tract (NTO) (STEPHAN, 1975; STEPHAN and ANDY, 1977). The border between the two components is characterized by a cell-poor lamina intermedia, by intercalated cell masses, and by different cell sizes and densities in the structures on either side. It remains recognizable even during the strong rotational changes which occur from Insectivora through Primates in connection with temporal lobe development.

This subdivision differs from the commonly used

classification, in that it combines the cortical nucleus with basal and lateral nuclei, rather than with central and medial nuclei in a corticomedial group (as first proposed by HUMPHREY, 1936). A cortico-basolateral combination was not new when proposed by STEPHAN (1975) and STEPHAN and ANDY (1977, Lit.): it had already been suggested, or at least traced out, by VÖLSCH (1911), JOHNSTON (1923), BERKELBACH VAN DER SPENKEL (1926), LOO (1931), and MACCHI (1951).

Of the finer parts of the amygdala, only the magnocellular part of the basal nucleus (MCB) and the nucleus of the lateral olfactory tract (NTO) could be delineated with sufficient accuracy for measurement.

Volume comparisons on the whole amygdaloid complex, on its two main components and on the two finer nuclei or subnuclei were performed by STEPHAN and ANDY (1977) in a small number of species (10 Insectivora, 1 tree-shrew, 7 prosimians, 8 simians including man) with the same methods as used here and in the earlier papers in this series (Parts I–VI). For the cortico-basolateral group, a clear progression from Insectivora through simians was found, whereas the centromedial group showed no clear changes. Since then, the number of species has been greatly enlarged and it is now possible to check the former results in the light of new morphological and experimental data.

Material and Methods

Volumes of the amygdaloid complex, its two main components (LAM, MAM) and two delimitable components (MCB, NTO) were measured in 89 species belonging to Macroscelidea (2 species), Insectivora (39 species), Scandentia (3 species), and Primates (18 species of prosimians and 27 species of simians) (Tables 1 and 2). More detailed information on the methods of (1) fixation and preparation of the brains and sections, (2) determination of volumes and their conversion to fresh values, and (3) interspecific and interstructural comparisons (percentage composition and least square regression analyses), are given in Parts I–VI of this series (FRAHM et al., 1982, 1984; STEPHAN et al., 1982, 1984; BARON et al., 1983, 1987) and by STEPHAN et al. (1981). In the last paper, body weights and the volumes of total (net) brains, telencephali and 'paleocortex + amygdala' are also given.

Regression line slopes close to 0.57 were found for the main olfactory bulb (BARON et al., 1983) and for most structures of the limbic system. The values of the various components of the amygdala also converge against this value; thus, this value is applied in the present paper for the reference lines of the structures (Figs. 11–12). The reference lines were drawn through the average of the values of the Insectivora ($N = 39$). Their y-intercepts are for

(1) amygdaloid complex	AMY =	0.332
(2) lateral amygdala	LAM =	0.066
(3) magnocell. basal ncl.	MCB =	-0.684
(4) medial amygdala	MAM =	-0.008
(5) ncl. lat. olf. tract	NTO =	-1.137
accessory olfactory bulb		AOB = -1.6745

Any value on the reference lines is considered to represent the average volume of the given structure for a typical Insectivora and has an index of 1 or 100(%). The distances from the reference line are expressed by deviations from 1 and area measure of the degree of deviation of the structural volumes from those of typical Insectivora. The indices are listed in Tables 1–3 and scaled in Figs. 13–14.

Results

Characteristics and delineation of structures

The positions of the various superficial parts of the amygdala and their rotational changes in connection with temporal lobe development from Insectivora through Primates are shown in Figs. 1–4. Superficial parts, i.e., those reaching the surface of the cerebral hemispheres, are present in the anterior amygdaloid area (AA), the nucleus tractus olfactorii lateralis (NTO), the ncl. medialis (M), and the nucleus corticis (CO), where several areas (COS, COP, COH) may be differentiated (Figs. 5–10). The superficial structures may as a whole be considered as periamygdaloid allocortex (STEPHAN, 1975, Lit.). More detailed descriptions appear in that paper and in STEPHAN and ANDY (1977). There are no superficial (= allocortical) components in nucleus centralis, basalis and lateralis.

In Nissl stained sections (Figs. 5–10), the border between centromedial and cortico-basolateral nuclear groups is generally clear and, in places, may be especially well marked by the lamina intermedia and their intercalated cell masses (I, e.g. in Fig. 8). The only exception is in Macroscelidea and especially *Rhynchocyon*, where the structural differentiation of most parts of the amygdala is poor. Only the nucleus tractus olfactorii lateralis (NTO) is well delimitable. Except for most simians the NTO is well differentiated (Fig. 5) and then clearly delimitable for volume measurements. The borders of the magnocellular part of the basal nucleus (MCB; Fig. 8) are distinct in most orders but less clear in Macroscelidea. The borders of the other nuclei within the two main groups are mostly uncertain and not precise enough to allow volume measurement.

Percentage composition and ratios

The percentages can easily be calculated from the volume data given in Tables 1 and 2. Average values are given in Table 3. We will point out some trends:

(1) The percentage of the amygdaloid complex distinctly increases from an average 21.7% in Insectivora to 58.5% in man (Table 3). The paleocortical complex complements these values, bringing them to 100 and, thus, undergoes a clear decrease (see Part VI of this series).

(2) In the amygdaloid complex, the percentage of the cortico-basolateral group distinctly increases from an average of 53.8% in Insectivora to 75.3% in man. The centromedial group complements these values, bringing them to 100 and, thus, undergoes a clear decrease.

(3) In the cortico-basolateral group, the large-celled part of the basal nucleus (MCB) decreases from an average of 17.9% in Insectivora to 8.8% in man. An even lower value (6.6%) is found in Macroscelidea, in which order the structural differentiation of MCB is poor. The small-celled parts complement these values, bringing them to 100 and, thus, undergo an increase.

(4) In the centromedial group, the nucleus of the lateral olfactory tract drops from an average of 7.4% in Insectivora to 0.07% in *Macaca mulatta*. In many non-human simians and in man we were not able to identify this nucleus with certainty in Nissl-stained sections. It may exist in these primates also, but would be very small and concealed in the relatively large MAM.

(5) The expected size of NTO in simians may be estimated by using the ratio NTO:BOL (Table 3; BOL data taken from Part VI). This ratio is on the average 1:75 in terrestrial Insectivora, 1:159 in prosimians, 1:294 in those simians in which NTO

Table 1. Volumes and indices of the amygdala (AMY) and amygdaloid components (LAM, MAM, MCB, NTO) in Macroscelidea, Insectivora and Scandentia

	Volumes in mm ³					Indices				
	AMY	LAM	MCB	MAM	NTO	AMY	LAM	MCB	MAM	NTO
MACROSCELIDEA										
<i>Elephantulus fuscipes</i>	35.6	19.7	1.3	15.9	0.91	165	169	64	161	125
<i>Rhynchocyon cirnei</i>	176.0	93.2	5.4	82.9	6.1	240	234	76	247	246
INSECTIVORA										
<i>Solenodon paradoxus</i>	133.8	83.1	14.1	50.7	2.5	129	148	141	107	71
<i>Tenrec ecaudatus</i>	88.2	49.1	9.1	39.1	3.1	89	91	95	86	92
<i>Setifer setosus</i>	68.7	36.6	8.1	34.5	2.9	138	136	169	141	170
<i>Hemicentetes semispinosus</i>	22.4	12.1	2.4	10.4	1.3	72	71	80	73	122
<i>Echinops telfairi</i>	22.4	11.3	3.2	11.1	0.70	82	76	120	89	75
<i>Oryzorictes talpoides</i>	20.8	11.1	1.2	9.7	0.64	111	110	67	114	101
<i>Microgale cowani</i>	14.8	9.3	0.91	5.5	0.70	146	170	93	119	202
<i>M. dobsoni</i>	17.0	10.2	1.7	6.8	0.75	108	120	114	95	142
<i>M. talazaci</i>	25.8	13.4	1.7	12.4	0.77	129	124	90	135	112
<i>Limnogale mergulus</i>	26.8	15.8	2.5	11.0	0.31	95	103	93	85	32
<i>Potamogale velox</i>	62.3	33.6	5.3	28.7	0.46	74	74	66	75	16
<i>Micropotamogale lamottei</i>	19.0	10.3	1.4	8.7	0.27	83	83	63	83	34
<i>M. ruwenzorii</i>	22.2	12.9	1.6	9.3	0.47	76	81	58	69	47
<i>Chrysochloris asiatica</i>	20.0	11.0	1.6	9.1	0.83	102	103	86	101	124
<i>C. stuhlmanni</i>	23.3	11.1	2.1	12.2	0.96	133	117	126	152	161
<i>Erinaceus algirus</i>	101.3	55.4	7.5	45.9	3.1	105	106	81	104	94
<i>E. europaeus</i>	108.1	63.3	11.5	44.8	3.7	107	116	118	97	109
<i>Hemiechinus auritus</i>	71.8	41.9	4.9	29.9	2.8	144	155	102	131	165
<i>Sorex araneus</i>	7.7	3.9	0.76	3.7	0.27	95	90	97	101	100
<i>S. cinereus</i>	4.9	2.5	0.54	2.4	0.24	96	91	110	102	141
<i>S. fumeus</i>	7.4	3.8	0.64	3.7	0.33	103	96	92	111	134
<i>S. minutus</i>	3.7	1.7	0.61	1.9	0.19	73	63	125	85	111
<i>Microsorex hoyi</i>	3.5	2.1	0.25	1.4	0.11	92	100	67	82	88
<i>Neomys fodiens</i>	8.6	4.4	1.2	4.1	0.24	84	80	118	88	69
<i>Blarina brevicauda</i>	10.9	5.9	0.89	5.0	0.45	93	93	79	93	114
<i>Crocidura flavescens</i>	13.0	6.6	1.2	6.4	0.43	82	77	81	88	79
<i>C. hildegardeae</i>	6.9	3.4	0.59	3.5	0.34	86	78	77	97	124
<i>C. jacksoni</i>	7.6	4.1	0.54	3.5	0.24	83	83	61	84	76
<i>C. russula</i>	6.3	3.2	0.73	3.1	0.23	75	71	90	80	80
<i>Suncus etruscus</i>	1.7	0.81	0.12	0.91	0.08	56	49	40	65	81
<i>S. murinus</i>	10.9	5.6	0.78	5.3	0.43	69	65	51	74	81
<i>Sylvisorex granti</i>	5.1	2.6	0.54	2.4	0.32	108	104	120	114	202
<i>S. megalura</i>	5.5	2.6	0.45	2.9	0.26	97	86	82	110	137
<i>Ruwenzorisorex suncoides</i>	10.2	5.5	0.69	4.7	0.18	91	90	64	92	47
<i>Myosorex babaulti</i>	12.1	5.9	1.1	6.2	0.58	112	100	105	126	159
<i>Scutisorex somereni</i>	17.8	9.8	1.7	8.0	0.79	78	79	75	77	101
<i>Desmana moschata</i>	90.8	49.9	12.2	40.9	0.09	132	134	183	130	4
<i>Galemys pyrenaicus</i>	36.5	21.5	5.7	15.0	0.23	169	183	272	151	31
<i>Talpa europaea</i>	26.3	15.0	3.5	11.3	0.56	104	109	143	98	65
SCANDENTIA										
<i>Tupaia glis</i>	87.3	59.3	4.6	28.0	2.2	218	272	120	153	158
<i>T. minor</i>	66.6	43.0	3.3	23.6	0.88	202	241	103	157	78
<i>Urogale everetti</i>	105.2	69.0	10.6	36.2	2.2	199	241	209	150	125

Table 2. Volumes and indices of the amygdala (AMY) and amygdaloid components (LAM, MAM, MCB, NTO) in Primates. (0 = not determinable with certainty)

	Volumes in mm ³					Indices				
	AMY	LAM	MCB	MAM	NTO	AMY	LAM	MCB	MAM	NTO
PROSIMIANS										
<i>Microcebus murinus</i>	36.6	24.5	2.1	12.0	0.30	175	217	103	126	42
<i>Cheirogaleus major</i>	102.3	68.8	8.8	33.6	0.95	146	182	130	105	40
<i>C. medius</i>	64.9	40.8	3.9	24.1	0.55	158	183	98	128	39
<i>Lemur albifrons</i>	260.7	185.5	20.0	75.2	1.5	195	257	156	123	33
<i>Varecia variegata</i>	368.2	253.5	22.9	114.7	2.6	179	227	115	122	37
<i>Lepilemur ruficaudatus</i>	133.9	91.4	7.8	42.5	0.64	128	161	77	89	18
<i>Avahi l. laniger</i>	113.2	73.8	5.4	39.5	1.1	89	107	44	68	26
<i>Avahi l. occidentalis</i>	121.1	86.5	6.4	34.6	0.27	120	158	66	75	8
<i>Propithecus verreauxi</i>	302.2	206.4	22.5	95.8	1.1	135	170	104	93	15
<i>Indri indri</i>	398.5	277.7	26.0	120.8	0.99	127	164	86	84	9
<i>Daubentonia madagasc.</i>	481.7	357.1	25.3	124.6	3.6	243	333	132	138	54
<i>Loris tardigradus</i>	104.1	72.5	6.0	31.6	0.47	180	232	108	120	24
<i>Nycticebus coucang</i>	189.2	121.4	11.4	67.8	1.1	195	231	122	153	32
<i>Perodicticus potto</i>	202.2	138.8	14.1	63.5	1.9	170	215	122	116	46
<i>Galago senegalensis</i>	73.2	51.5	4.2	21.7	0.55	173	225	104	113	39
<i>Otolemur crassicaudatus</i>	148.8	103.3	8.5	45.5	1.4	148	190	88	99	40
<i>Galagoides demidoff</i>	57.9	40.4	3.2	17.6	0.73	220	283	127	146	82
<i>Tarsius spp.</i>	53.4	39.7	3.3	13.7	0.09	159	217	102	89	8
SIMIANS										
<i>Callithrix jacchus</i>	106.1	75.6	7.7	30.5	0	199	262	149	125	0
<i>Cebuella pygmaea</i>	75.4	57.9	7.1	17.5	0.11	229	325	225	116	10
<i>Saguinus midas tamarin</i>	144.9	106.1	8.8	38.8	0	243	329	153	143	0
<i>S. oedipus</i>	141.8	110.2	11.4	31.6	0.12	224	320	186	109	5
<i>Callimico goeldii</i>	146.9	111.2	11.6	35.6	0	203	283	167	108	0
<i>Cebus albifrons</i>	458.2	337.8	36.0	120.4	0	218	297	178	126	0
<i>Aotus trivirgatus</i>	193.6	136.9	14.6	56.6	0.21	195	255	153	125	6
<i>Callicebus moloch</i>	254.1	179.5	16.3	74.6	0	245	319	163	157	0
<i>Saimiri sciureus</i>	242.4	188.3	15.5	54.1	0	279	400	186	136	0
<i>Pithecia monacha</i>	365.3	273.7	31.7	91.6	0	263	364	237	144	0
<i>Alouatta seniculus</i>	426.0	324.3	31.4	101.7	0	134	189	103	70	0
<i>Ateles geoffroyi</i>	868.8	644.5	74.7	224.3	0	241	330	215	136	0
<i>Lagothrix lagothricha</i>	753.2	565.2	74.6	188.0	0	267	370	274	146	0
<i>Macaca mulatta</i>	678.4	493.2	69.7	185.2	0.13	191	256	203	114	1
<i>Cercocebus albigena</i>	781.3	515.9	83.5	265.4	0	218	266	242	162	0
<i>Papio anubis</i>	953.5	740.2	104.7	213.3	0	138	198	157	68	0
<i>Cercopithecus ascanius</i>	572.5	433.2	45.6	139.3	0	259	361	214	138	0
<i>C. mitis</i>	704.0	493.1	66.4	210.9	0	224	289	219	147	0
<i>Miopithecus talapoin</i>	413.1	311.3	52.9	101.9	0	338	470	449	182	0
<i>Erythrocebus patas</i>	688.4	489.0	63.7	199.4	0	194	254	186	123	0
<i>Colobus badius</i>	500.9	377.3	47.7	123.6	0	150	208	148	81	0
<i>Pygathrix nemaeus</i>	489.2	337.5	51.2	151.6	0	141	179	153	95	0
<i>Nasalis larvatus</i>	718.9	499.1	55.8	219.8	0	145	186	117	97	0
<i>Hylobates lar</i>	666.2	510.2	84.6	156.0	0	224	317	296	115	0
<i>Pan troglodytes</i>	1421.8	1046.5	136.0	375.3	0	146	198	144	84	0
<i>Gorilla gorilla</i>	2752.6	1998.1	227.5	754.5	0	176	236	151	106	0
<i>Homo sapiens</i>	5286.6	3980.2	349.9	1306.4	0	444	617	305	240	0

Table 3. Indices and percentages (means of various taxonomic and ecoethological groups). BAC = accessory olfactory bulb; BOL = main olfactory bulb; P&A = paleocortex + amygdala; N = number of species; *) only those species included in which NTO could be identified (for N see Table 2).

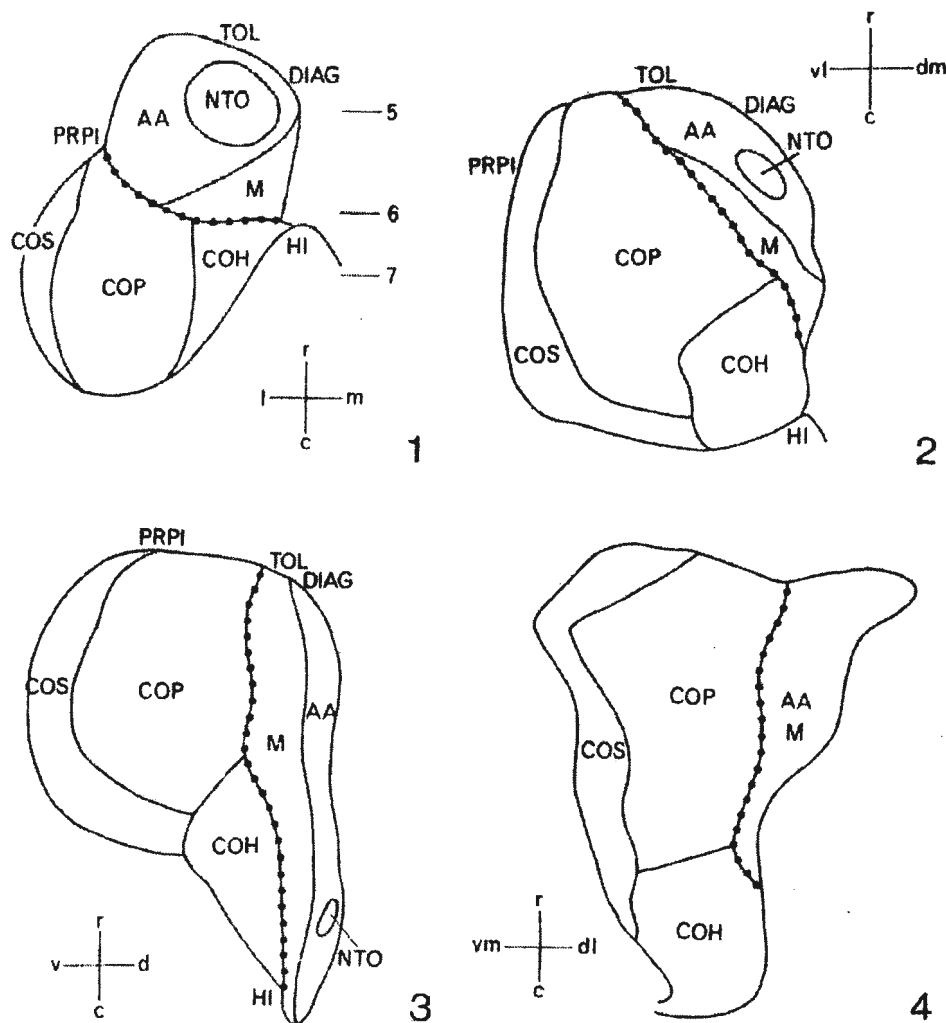
	Indices								Percentages				Ratios
	N	BOL	BAC	AMY	LAM	MCB	MAM	NTO	AMY	LAM	MCB	NTO	NTO:BOL
									P&A	AMY	LAM	MAM	
MACROSCELIDEA	2	174	202	203	202	70	204	185	27.3	54.2	6.3	6.6	1 : 70
INSECTIVORA	39	100	100	100	100	100	100	100	21.7	53.8	17.9	7.4	83
Terrestrial species	32	109	102	100	99	95	101	114	20.0	53.4	17.6	8.4	75
Semiaquatic species	7	60	93	102	105	122	97	33	29.4	55.8	19.3	2.9	126
Solenodontidae	1	178	21	129	148	141	107	71	19.4	62.1	17.0	4.9	186
Tenrecinae	4	129	85	95	93	116	97	115	18.5	53.3	22.2	8.9	87
Oryzorictinae	5	117	129	118	125	92	109	118	21.9	57.5	13.3	7.9	86
Potamogalinae	3	48	101	78	79	62	76	32	29.3	55.4	14.0	3.2	126
Chrysochloridae	2	127	93	117	110	105	126	143	17.6	51.1	17.0	8.5	67
Erinaceidae	3	129	230	119	126	100	111	122	23.1	57.2	14.5	8.1	81
Soricidae	18	89	82	87	83	85	93	107	20.2	51.4	18.5	8.5	65
Talpidae	3	82	83	135	142	199	126	34	28.7	57.0	24.7	2.2	137
SCANDENTIA	3	128	561	206	252	144	153	120	31.4	66.0	10.3	5.9	84
PROSIMIANS	18	67	371	163	208	105	110	33	39.6	68.9	9.0	2.1	159
Cheirogaleidae	3	90	681	160	194	110	120	40	33.0	65.7	10.2	2.5	165
Lemurinae	2	66	252	187	242	136	123	35	39.0	70.0	9.9	2.1	140
Lepilemurinae	1	43	265	128	161	77	89	18	44.3	68.3	8.5	1.5	177
Indridae	4	26	185	118	150	75	80	14	44.6	68.6	8.7	1.4	160
Daubentonidae	1	137	426	243	333	132	138	54	31.3	74.1	7.1	2.9	189
Lorisidae	6	82	408	181	229	112	125	44	37.3	68.6	8.7	2.6	146
Tarsiidae	1	22	249	159	217	102	89	8	57.6	74.3	8.3	0.7	202
SIMIANS	27	10	27	220	299	199	126	1	54.3	73.6	12.0	0.05	294 *)
Callitrichidae	5	14	68	220	304	176	120	3	58.0	74.9	10.3	0.2	134 *)
Cebidae	8	11	50	230	315	189	130	1	53.3	74.1	10.6	0.04	269 *)
Cercopithecinae	7	12	0	223	299	239	133	0.1	52.6	72.6	14.1	0.01	638 *)
Colobinae	3	3	0	145	191	139	91	0	54.0	71.3	13.0	0	-
Pongidae	3	8	0	182	250	197	102	0	53.6	74.3	13.7	0	-
Hominidae	1	4	0	444	617	305	240	0	58.5	75.3	8.8	0	-
MACROSCELIDEA	2	174	202	203	202	70	204	185	27.3	54.2	6.3	6.6	1 : 70
INSECTIVORA	39	100	100	100	100	100	100	100	21.7	53.8	17.9	7.4	83
SCANDENTIA	3	128	561	206	252	144	153	120	31.4	66.0	10.3	5.9	84
PROSIMIANS	18	67	371	163	208	105	110	33	39.6	68.9	9.0	2.1	159
NON-HUMAN SIMIANS	26	11	28	211	287	195	121	1	54.1	73.6	12.1	0.05	294 *)
MAN	1	4	0	444	617	305	240	0	58.5	75.3	8.8	0	-

could be identified, and 1 : 638 in *Macaca*, which is the only Old World monkey in which we were able to identify this nucleus. Using the ratio of simians to estimate NTO volume in man gives a value of about 0.4 mm^3 ($113.5/294$; 113.5 mm^3 is the BOL volume in man; see BARON et al., 1987), whereas if *Macaca* is used it would be about 0.2 mm^3 ($113.5/638$). A structure of this size may be hardly identifiable in man, in which the MAM as a whole is 3–6 thousand times larger (1306 mm^3 ; see Table 2).

Size indices

Total amygdaloid complex (AMY).

The size indices of the amygdaloid complex based on the average Insectivora are given in Tables 1 and 2, and averaged for supraspecific taxonomic units in Table 3. From Insectivora through prosimians and non-human simians to man, there is a clear increase in relative size (1 : 1.6 : 2.1 : 4.4). Man had the largest relative size of amygdala. In Insectivora no differen-



Figs. 1–4. Reconstructions of the surface of the amygdala of the long-eared hedgehog (*Hamiechinus auritus*) (Fig. 1; right hemisphere from ventral), the Indri (*Indri indri*) (Fig. 2; right hemisphere from ventromedial), the squirrel monkey (*Saimiri sciureus*) (Fig. 3; right hemisphere from medial; based on the atlas of EMMERS and AKERT, 1963), and man (*Homo sapiens*) (Fig. 4; right hemisphere from dorsomedial; based on BROCKHAUS, 1940). The superficial structures may be considered as periamygdaloid allocortex. The border between cortico-basolateral and centromedial amygdaloid main groups is marked by large dots. Subdivisions within these main groups are based on STEPHAN (1975).

AA anterior amygdaloid area, COH area parahippocampalis of cortical nucleus, COP area principalis of cortical nucleus, COS area semiannularis of cortical nucleus, DIAG diagonal band of Broca, HI hippocampus, M nucleus medialis, NTO nucleus of the lateral olfactory tract, PRPI prepiriform cortex, TOL olfactory tubercle. — Determinants of direction: c caudal, d dorsal, l lateral, m medial, r rostral, v ventral; they may also be combined. The numbers 5–7 indicate the approximate positions of Figs. 5–7.

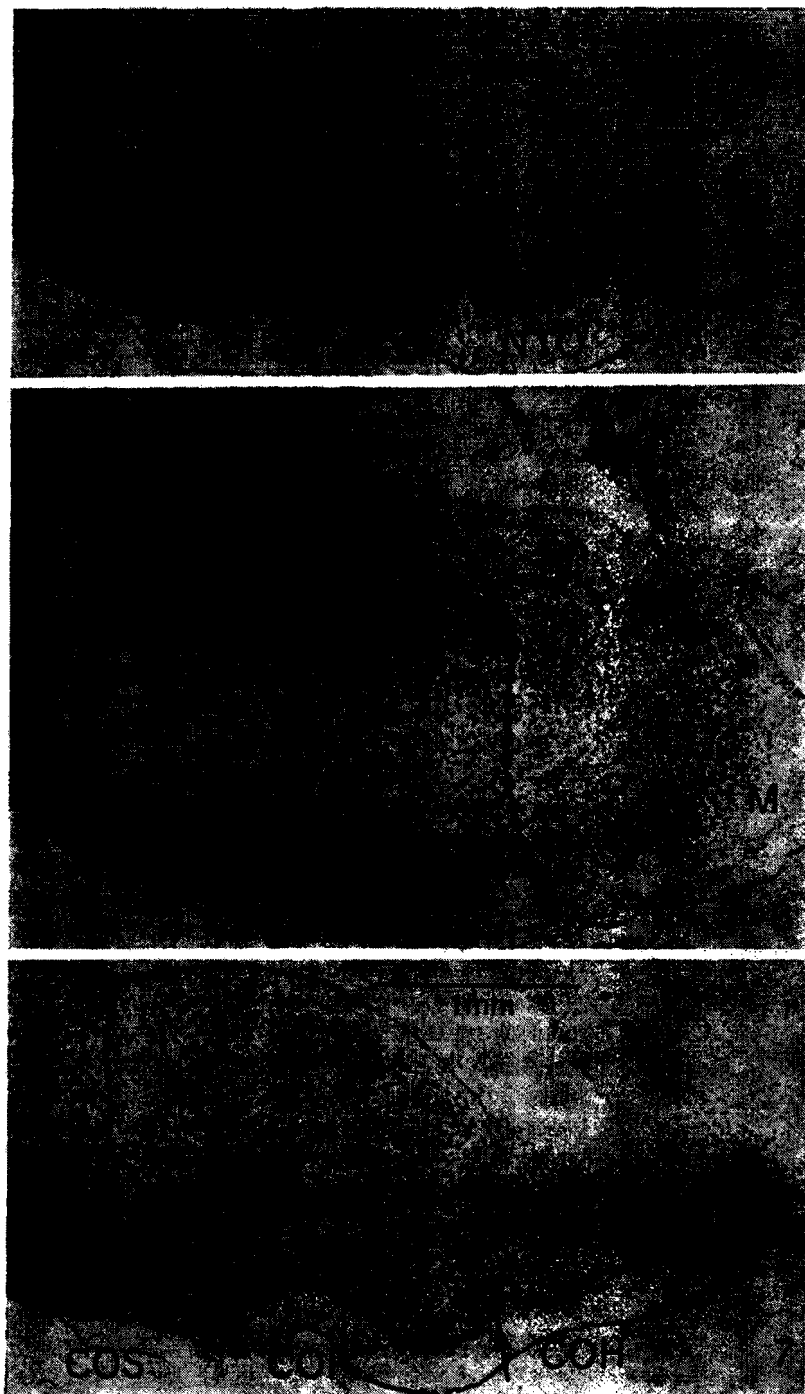
ces were found between terrestrial and water-adapted forms. In Macroscelidea and Scandentia, the amygdala was twice as large as in Insectivora.

Cortico-basolateral amygdaloid group (LAM)

The size indices of the LAM (Fig. 13) undergo a much greater increase than those of the total amygdala (AMY): in all primate species, the LAM had higher indices than the AMY. From Insectivora through prosimians and non-human simians to man, the average indices of the LAM are 1 : 2.1 : 2.9 : 6.2. In Insec-

tivora, those families and/or subfamilies with the lowest indices for the AMY had similarly low or even lower values for the LAM; those with the highest indices for the AMY had even higher values for the LAM.

Within the LAM, the large-celled basal nucleus (MCB) is distinctly less progressive than the smaller-celled parts. In many primates the indices of the MCB are on the average only half that of the LAM. Consequently, the small-celled components increase to a greater extent than the LAM as a whole. Of



Figs. 5–7. Frontal sections through the amygdaloid complex of the long-eared hedgehog (*Hemiechinus auritus*). Cresyl-violet stain, enlargement $27.6\times$. The position of the sections is marked in Fig. 1. The arrows indicate periamygdaloid subdivisions of the cortical nucleus. For abbreviations see Figs. 1–4. Further abbreviations for Figs. 5–10: B nucleus basalis, CE nucleus centralis, E entorhinal cortex, I interstitial cell masses, L nucleus lateralis, MCB magnocellular part of basal nucleus, STR striatum.

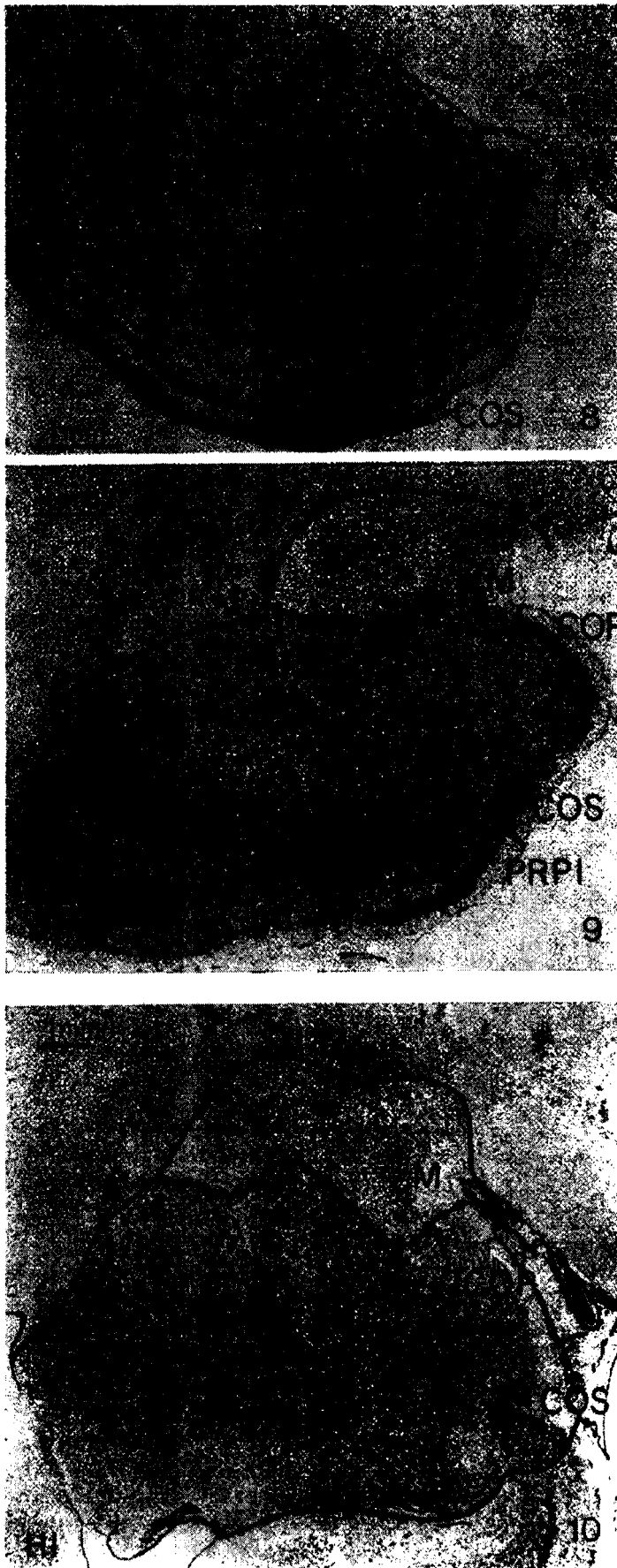
them, the lateral nucleus seems to be the most progressive. This statement, however, is based solely on impressions obtained from photoprints and could not be verified by measurement.

Centromedial amygdaloid group (MAM).

From Insectivora through prosimians and non-human simians to man, the average indices of the MAM (Fig. 14) attain clearly higher values only in man. The average indices are 1 : 1.1 : 1.2 : 2.4. Also within the

orders and suborders the differences are relatively small, i.e., MAM is relatively stable.

Within the MAM, the nucleus of the lateral olfactory tract (NTO) is much more variable and behaves quite differently. Its average indices are (1) much smaller in semiaquatic than in terrestrial Insectivora ($33 : 114$, i.e., less than $1/3$), (2) much smaller in prosimians than in Insectivora on the average ($33 : 100$) and (3) extremely small in simians ($1 : 100$). In many simians and in man, we were not able to identi-



fy NTO in the MAM complex with certainty. Its expected volume in man would be about 0.4 mm^3 if the same index as found for *Macaca* is adopted. Similar volumes were estimated with reference to the NTO : BOL ratios found in non-human simians (see percentage composition and ratios).

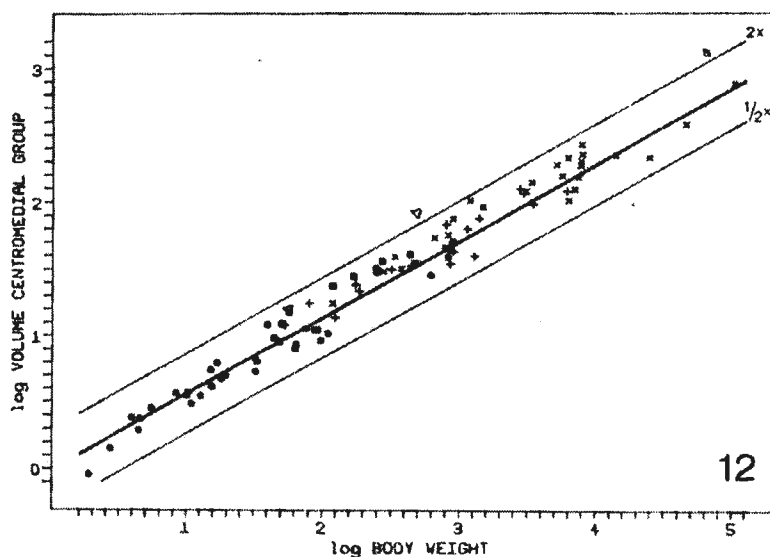
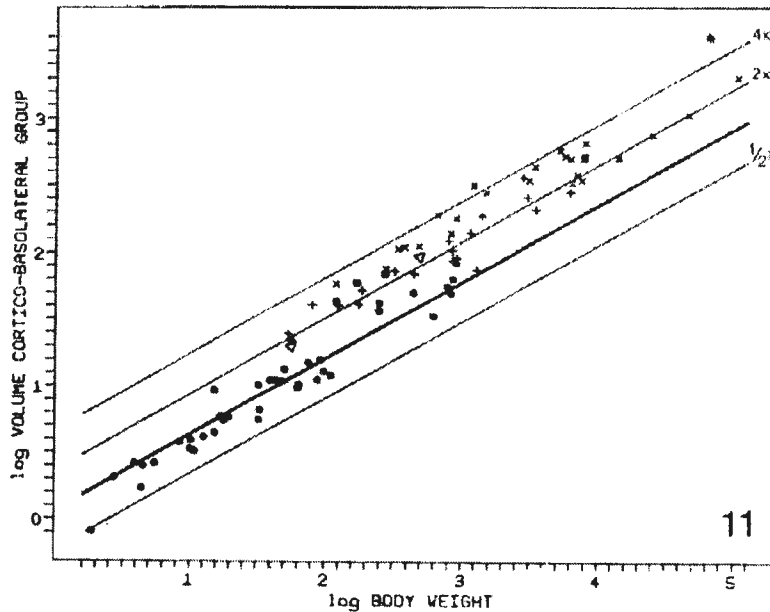
When the average indices of NTO are compared with the average indices of BOL they are 114 : 109 in terrestrial Insectivora (see Table 3), 33 : 60 in semiaquatic Insectivora, 33 : 67 in prosimians, and 5.5 : 14 in simians with recognizable NTO ($N = 4$). Some prosimian species have relatively large BOL (indices > 100 ; *Daubentonia*, *Galagoides*, *Perodicticus*; see Part VI), while others have relatively small BOL (indices < 30 ; all Indriidae, *Tarsius*). In these extreme groups, the corresponding ratios (NTO index : BOL index) are 61 : 122 in the macrosomatic forms ($N = 3$) and 13 : 25 in the microsomatic forms ($N = 5$). If the BOL indices are uniformly set at 1, the average NTO index for terrestrial Insectivora is 1.05 times larger than the average BOL index, but the NTO indices in all other groups compared above are distinctly smaller: 0.55 times for semiaquatic Insectivora, 0.49 times for all prosimians, 0.50 times for macrosomatic prosimians, 0.52 times for microsomatic prosimians, and 0.39 times for the four simian species with recognizable NTO. When comparing the macrosomatic prosimians with the macrosomatic (terrestrial) Insectivora it is of interest that the NTO indices of the former are much smaller (61 versus 114) in spite of larger BOL indices (122 versus 109). Thus, NTO is not only reduced when BOL is reduced, but, in primates, it can be reduced with no BOL reduction. In primates, the importance of olfaction appears to be generally decreased in that functional complex to which NTO is considered to mediate olfactory input; olfaction appears to be partly replaced by other senses (see Discussion).

Discussion

Subdivisions of the amygdala

The subdivision of the amygdala used in this paper was substantiated by STEPHAN (1975) and STEPHAN and ANDY (1977, Lit.). In 1979, TEN DONKELAAR et al. pointed out that their results on the time of neuron origin in the amygdaloid nuclear complex of the Chinese hamster confirms this subdivision with

Figs. 8–10. Frontal sections through the amygdaloid complex of the sportive lemur (*Lepilemur ruficaudatus*) (Fig. 8), the howler monkey (*Alouatta* sp.) (Fig. 9), and man (*Homo sapiens*) (Fig. 10). Cresylviolet stain, enlargement as indicated. For abbreviations see Figs. 1–4 and 5–7.



Figs. 11–12: Volumes (in mm³) of cortico-basolateral (Fig. 11) and centromedial (Fig. 12) amygdaloid groups plotted against body weight (in grams) in double logarithmic scales. The thick reference lines are through the averages of the 39 species of Insectivora so far measured (see Material and Methods) and have the equations

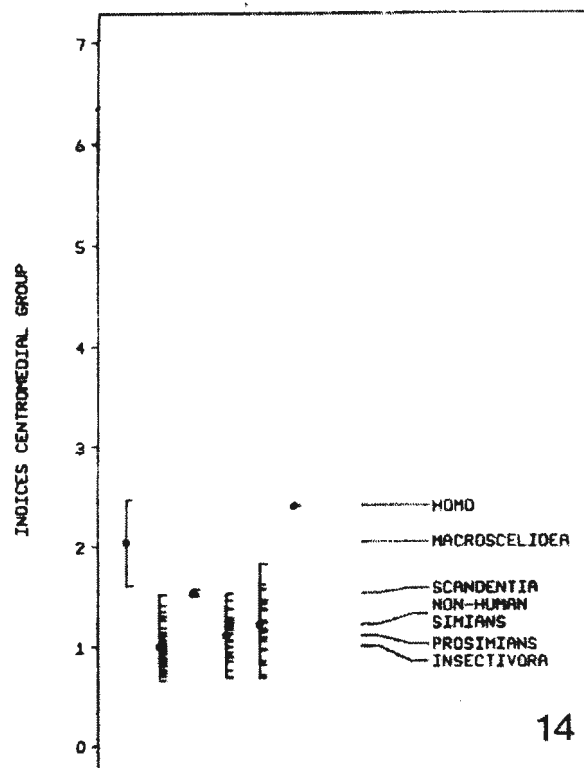
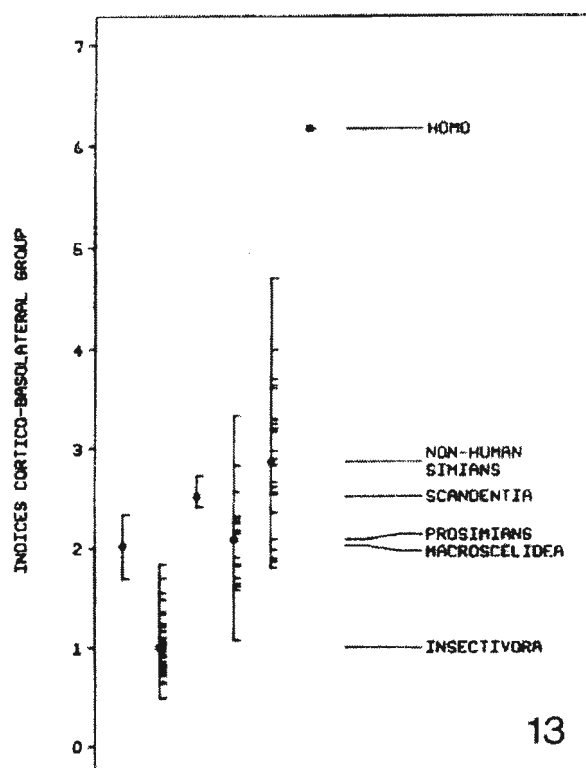
$$\log \text{LAM volume} = 0.066 + 0.57 \times \log \text{body weight},$$

$$\log \text{MAM volume} = -0.008 + 0.57 \times \log \text{body weight}.$$

The thinner parallel lines are in multiples of 2, i.e., two or four times, or 1/2 time, respectively, and refer to the size indices ($\times 100$) given in Tables 1–3 and scaled in Figs. 13–15. ∇ Macroscelidea, $\bullet$ Insectivora, $\blacksquare$ Scandentia (tree-shrews), $\pm$ prosimians, $\times$ non-human simians, $\bullet$ man.

the exception of the 'anterior cortical nucleus'. According to STEPHAN (1975), however, the concerned anterior superficial parts of the amygdala are inseparable parts of the centromedial complex (anterior amygdaloid area; AA in Figs. 1–4), and clearly separated from the cortical nucleus. With due regard of the differences in the classification of these anterior parts the distinction between two groups of nuclei in the amygdaloid complex proposed by TEN DONKELAAR et al. (1979) is exactly the same as the subdivision proposed by STEPHAN (1975) and STEPHAN and ANDY (1977). Finally, data on the neurogenesis in the rat amygdala (BAYER, 1980) can be interpreted in the same manner, although that author does not discuss the problem.

Histochemical methods may lead to similar divisions of the amygdaloid complex. Many of them, however, (1) reveal a relatively high degree of heterogeneity within the nuclei (superficial areas included), (2) result in delimitations going across nuclei and thus (3) may point to divisions which differ from those based on cytoarchitecture. A survey of chemoarchitecture, radioimmunoassay and immunohistochemical techniques, as e.g. based on studies of HALL and GENESER-JENSEN (1971, acetylcholinesterase, monoaminooxidase), HALL et al. (1969, dithizone, Timm stains), BEN-ARI et al. (1975, 1976, 1977, norepinephrine, dopamine, GABA, GAD, HD), and EMSON et al. (1978, substance P) suggests that the principal nuclei contain subdivisions with marked differences in their



Figs. 13–14. Indices of the cortico-basolateral (Fig. 13) and centromedial (Fig. 14) amygdaloid groups. The indices are numerical values of the distances from the thick reference baseline in Figs. 11–12 and are given in their percentage form ($\times 100$). Each of the small horizontal bars represents the index for a species. On the left, the ranges (vertical bars) and the means (o) are given, while on the right, the sequence of the means is scaled and the names of the units are given.

functional properties. There is also considerable heterogeneity in the fiber connections, as discussed below.

Functional aspects of size differences

The amygdala, which may be considered part of the limbic system (STEPHAN, 1983; Lit.), plays an important role not only in the control of emotional functions through its connections with the hypothalamus, but also has an influence on cortical cognitive function (MISHKIN and AGGLETON, 1981). This role as a link between hypothalamus and cortex requires that the amygdala be a major site of convergence for vegetative and sensory information. There is now sufficient anatomical evidence that the amygdala does indeed receive input from various modalities and projects to many functionally different brain centers, including the neocortex.

The question arises as to what extent the size increase of the cortex during evolution, reflecting an increase in cognitive functions, is accompanied by a size increase of the amygdala. The fact that changes in the limbic structures occurred during evolution was already established by quantitative investigations of brain structures in Insectivora and Primates

(STEPHAN and ANDY, 1964). An early detailed investigation on quantitative changes of components of the amygdaloid complex by STEPHAN and ANDY (1977) revealed the existence of significant differences in the relative size of the nuclear groups among Insectivora and Primates.

The cortico-basolateral group (LAM).

LAM is larger in Primates than in Insectivora, and increases strongly in man. The marked increase indicates that its functional capacities, as indicated by experimental-anatomical, electrophysiological and behavioral studies, become more predominant inside the amygdala. The increase is generally stronger in the small-celled components of this nuclear group and parallels to a certain extent the extension of neocortex from primitive Insectivora to Primates and man (STEPHAN and ANDY, 1964; STEPHAN et al., 1987).

The most important, though not the exclusive input to this region of the amygdala is from modality specific association and polysensory convergence areas, particularly of the temporal cortex or homologous regions; other cortical areas that have been suggested are the frontal lobe and the insular cortex (rat: OTTERSEN, 1981, 1982; cat: HIRATA, 1965;

DRUGA, 1969/70; RUSSCHEN, 1982; *rhesus monkey*: WHITLOCK and NAUTA, 1956; HERZOG and VAN HOESEN, 1976; AGGLETON et al., 1980; TURNER et al., 1980; VAN HOESEN, 1981). The neocortical-amygdaloid link has also been demonstrated by electrophysiological studies (NIEMER and GOODFELLOW, 1966; PRELEVIĆ et al., 1976). Several behavioral studies also indicate an important role, especially of temporal neocortical inputs to the amygdala, for the perception and interpretation of environmental stimuli (PRIBRAM and BAGSHAW, 1953; AKERT et al., 1961; HOREL and KEATING, 1969; JONES and MISHKIN, 1972; MYERS, 1972; HOREL and MISANTONE, 1974).

The increase of the LAM group in the Insectivora-Primates line strongly suggests that the amygdala of the simians, and particularly of man, is much more influenced by sensory neocortical areas than the amygdala of the Insectivora, which may be more influenced by olfaction. Since the cortical nucleus receives fibers from the main olfactory bulb, its size may be expected to regress with the marked reduction of the bulb in semiaquatic Insectivora and Primates (see Part VI). No well-founded numerical statements could be made, since the borders between cortical and basal nucleus are indistinct and thus, these nuclei are not amenable to precise measurement. The impression obtained from microscopic examination of sections and from photoprints was not that there had been a reduction, but the cortical nucleus as a whole appears to be well developed in primates (Figs. 8–10) in which the olfactory bulb is strongly reduced. Structural alterations similar to those described for the olfactory tubercle (see Part VI) were not found. Overall, there is reason to doubt that all parts combined under the term 'cortical nucleus' are related to olfaction.

The dorsomedial part of the lateral nucleus seems to be the site of convergence of all five sensory modalities (VAN HOESEN, 1981). In this context, it is of interest that the lateral nucleus appears to be the most progressive of all amygdaloid components. The amygdala, and particularly its LAM group, is the pivotal structure in the interaction between sensory information and drives, on the one hand, and between affect and cognition on the other (see e.g., MISHKIN and AGGLETON, 1981). Any influence that the neocortical sensory systems have on emotional processes, and any influence that the visceral system has on sensory and cognitive processes are mediated by connections in the amygdala and, to a large extent, in the nuclei of the LAM. Therefore, the functional role of the LAM increases with the amount of environmental information which can be processed by the neocortex, or in other words with the complexity of the ecological niche. However, ecological descriptors are

not yet available in sufficient number and precision to show significant correlations between parts of the amygdala and the complexity of the ecological niche.

The centromedial amygdaloid group (MAM).

The size decrease of the nucleus of the lateral olfactory tract (NTO) from Insectivora to simians is more accentuated than that of the BOL. NTO may still be present but it is no more identifiable with certainty in many simians and in man. In man its expected size may be at the most 0.4 mm³.

Based on fiber connections, SHAMMAH-LAGNADO and NEGRAO (1981) have suggested that the NTO could constitute a link between the main olfactory and the accessory olfactory (vomeronasal) systems. However, relative size and differentiation of the accessory bulb (AOB, STEPHAN, 1965; STEPHAN et al., 1982) is without recognizable influence on the NTO indices. This may be because of the very small size of the AOB, which is on the average only 1.2% of the main olfactory bulb and at most 4.4% (in *Tarsius*). The degree of reduction in NTO depends primarily on the main olfactory system but it is not parallel to that of BOL (see Results: Size indices). In macrosomatic representatives of prosimian primates, NTO becomes reduced with no BOL reduction, and in semiaquatic Insectivora and microsmatic representatives of primates, NTO reduction is in general much stronger than BOL reduction. This may indicate that NTO is a link for olfactory input into a polysensory system which may be concerned with the search for food. Olfaction may become useless or of less use for the given functional complex and be replaced by non-olfactory senses, e.g., trigeminal innervated vibrissae in water adapted Insectivora, or vision in primates.

The central nucleus, which receives widespread intrinsic afferents (AGGLETON, 1985) mainly from nuclei which are targets of cortico-amygdaloid projections (HERZOG and VAN HOESEN, 1976; AGGLETON et al., 1980; TURNER et al., 1980), sends fibers to and receives fibers from a variety of nuclei dispersed over the whole brainstem (HOPKINS and HOLSTEGE, 1978; OTTERSEN and BEN-ARI, 1978; PRICE and AMARAL, 1981; SCHWABER et al., 1982; CECETTO et al., 1983; TAKEUCHI et al., 1983). Because of its fiber connections, the central nucleus may be considered a link between neocortical sensory association areas and autonomic centers of the brainstem. These projections support the notion of an involvement by the amygdala in control of the visceral, and particularly the cardiovascular, components in emotional behavior (SCHWABER et al., 1982).

Based on studies in rodents, the medial nucleus of the MAM seems to be mainly related to the vomeronasal system by its reciprocal connections with the

accessory olfactory bulb (DE OLMOS and INGRAM, 1972; RAISMAN, 1972; BROADWELL, 1975; SCALIA and WINANS, 1975; KRETTEK and PRICE, 1978; KEVETTER and WINANS, 1981) but it seems to receive olfactory input as well (LEHMAN and WINANS, 1982), at least by intraamygdaloid projections (KRETTEK and PRICE, 1978; KEVETTER and WINANS, 1981). The most important efferents of the medial nucleus project to hypothalamic areas (HEIMER and NAUTA, 1969; DE OLMOS and INGRAM, 1972; KRETTEK and PRICE, 1978; KEVETTER and WINANS, 1981) which have been shown to be implicated in the control of reproductive physiology and behavior (MALSURY and PFAFF, 1974; LARSSON, 1979).

Since the medial nucleus receives fibers from the accessory olfactory system, one would expect the size of this structure to regress with the disappearance of the vomeronasal organ. This seems, however, not to be the case since the medial nucleus exists in Old World simians, including man (STEPHAN, 1975; STEPHAN and ANDY, 1977) that apparently lack an accessory olfactory system (STEPHAN et al., 1982). According to STEPHAN (1975), the structural pattern of the medial nucleus does not change in the prosimian-simian line and is also well differentiated in man.

In evaluating the functional significance of the size of the medial nucleus one should not ignore its intraamygdaloid connections. In fact, in primates, the medial nucleus receives major fiber projections from the lateral, basal, and accessory basal nuclei (AGGLETON, 1985) as is also the case in the central nucleus. Thus, the medial nucleus may represent an important relay for complex sensory information running to areas implicated in the control of reproductive physiology and behavior.

While the LAM group is best characterized by its connections with sensory systems in the neocortex and the paleocortex (though other connections, particularly the intraamygdaloid projections, are also important), the MAM group is characterized mainly by its connections with brainstem nuclei. The fact that the MAM group did not undergo as drastic a size increase as the LAM group can be easily explained by its function as revealed by the respective fiber connections discussed above. The visceral system, being the major source and target of connections with the MAM, may be considered conservative during evolution. Visceral functions are essentially the same in primitive and in higher mammals. The somatic system, related at the amygdaloid level mainly to the LAM, however, is characterized by a progressive evolution enabling organisms to engage in increasingly complex interactions with the environment.

Acknowledgements

The excellent assistance of Cläre ROBERG, Monika MARTIN, Sigrid WADLE, Regina KRAUSS (histology), Helga GROBECKER (photography), Michael STEPHAN (computer work), and Helma LEHMANN (secretarial work) is gratefully acknowledged. Thanks are due to Jessica POTTIER (Montréal) for checking the English text.

References

- AGGLETON, J. P.: A description of intra-amygdaloid connections in old world monkeys. *Exp. Brain Res.* **57**, 390—399 (1985).
- AGGLETON, J. P., BURTON, M. J. and PASSINGHAM, R. E.: Cortical and subcortical afferents to the amygdala of the rhesus monkey (*Macaca mulatta*). *Brain Res.* **190**, 347—368 (1980).
- AKERT, K., GRUESSEN, R. A., WOOLSEY, C. N. and MEYER, D. R.: Kluver-Bucy syndrome in monkeys with neocortical ablations of temporal lobe. *Brain* **84**, 480—498 (1961).
- BARON, G., FRAHM, H. D., BHATNAGAR, K. P. and STEPHAN, H.: Comparison of brain structure volumes in Insectivora and Primates. III. Main olfactory bulb (MOB). *J. Hirnforsch.* **24**, 551—568 (1983).
- BARON, G., STEPHAN, H. and FRAHM, H. D.: Comparison of brain structure volumes in Insectivora and Primates. VI. Paleocortical components. *J. Hirnforsch.* **28**, 463—477 (1987).
- BAYER, S. A.: Quantitative ³H-thymidine radiographic analyses of neurogenesis in the rat amygdala. *J. Comp. Neurol.* **194**, 845—876 (1980).
- BEN-ARI, Y., KANAZAWA, I. and ZIGMOND, R. E.: Regional distribution of glutamate decarboxylase and GABA within the amygdaloid complex and stria terminalis system of the rat. *J. Neurochem.* **26**, 1279—1283 (1976).
- BEN-ARI, Y., LE GAL LA SALLE, G., BARDIN, G., SCHWARTZ, J. C. and GARBARG, M.: Histamine synthesizing afferents within the amygdaloid complex and bed nucleus of the stria terminalis of the rat. *Brain Res.* **138**, 285—294 (1977).
- BEN-ARI, Y., ZIGMOND, R. E. and MOORE, K. E.: Regional distribution of tyrosine hydroxylase, norepinephrine and dopamine within the amygdaloid complex of the rat. *Brain Res.* **87**, 96—101 (1975).
- BERKELBACH VAN DER SPENKEL, H.: Stria terminalis and amygdala in the brain of the opossum (*Didelphis virginiana*). *J. Comp. Neurol.* **42**, 211—254 (1926).
- BROADWELL, R. D.: Olfactory relationships of the telencephalon and diencephalon in the rabbit. I. An autoradiographic study of the efferent connections of the main and accessory olfactory bulbs. *J. Comp. Neurol.* **163**, 329—345 (1975).
- BROCKHAUS, H.: Zur normalen und pathologischen Anatomie des Mandelkerngebietes. *J. Psychol. Neurol. (Leipzig)* **49**, 1—136 (1940).
- CECHETTO, D. F., CIRIELLO, J. and CALARESCU, F. R.: Afferent connections to cardiovascular sites in the amygdala: a horseradish peroxidase study in the cat. *J. Auton. Nerv. Syst.* **8**, 97—110 (1983).
- DE OLMOS, S. J. and INGRAM, W. R.: The projection field of the stria terminalis in the rat brain. An experimental study. *J. Comp. Neurol.* **146**, 303—334 (1972).
- DRUGA, R.: Neocortical projections to the amygdala (An experimental study with the Nauta method). *J. Hirnforsch.* **11**, 467—476 (1969/70).
- EMMERS, R. and AKERT, K.: A Stereotaxic Atlas of the Brain of the Squirrel Monkey (*Saimiri sciureus*). Univ. of Wisconsin Press, Madison (1963).

- EMSON, P. C., JESSELL, T., PAXINOS, G. and CUELLO, A. C.: Substance P in the amygdaloid complex, bed nucleus and stria terminalis of the rat brain. *Brain Res.* **149**, 97–105 (1978).
- FRAHM, H. D., STEPHAN, H. and BARON, G.: Comparison of brain structure volumes in Insectivora and Primates. V. Area striata. *J. Hirnforsch.* **25**, 537–557 (1984).
- FRAHM, H. D., STEPHAN, H. and STEPHAN, M.: Comparison of brain structure volumes in Insectivora and Primates. I. Neocortex. *J. Hirnforsch.* **23**, 375–389 (1982).
- HALL, E. and GENESER-JENSEN, F. A.: Distribution of acetylcholinesterase and monoamine oxidase in the amygdala of the guinea pig. *Z. Zellforsch.* **120**, 204–221 (1971).
- HALL, E., HAUG, F. M. S. and URSIN, H.: Dithizone and sulphide silver staining of the amygdala in the cat. *Z. Zellforsch.* **102**, 40–48 (1969).
- HEIMER, L. and NAUTA, W. J. H.: The hypothalamic distribution of the stria terminalis in rat. *Brain Res.* **13**, 284–297 (1969).
- HERZOG, A. G. and VAN HOESEN, G. W.: Temporal neocortical afferent connections to the amygdala in the rhesus monkey. *Brain Res.* **115**, 57–69 (1976).
- HIRATA, Y.: Subcortical projections from the orbital surface of the cat's brain. *Acta med. et biol.* **13**, 123–142 (1965).
- HOPKINS, D. A. and HOLSTEGE, G.: Amygdaloid projections to the mesencephalon, pons and medulla oblongata in the cat. *Exp. Brain Res.* **32**, 529–547 (1978).
- HOREL, J. A. and KEATING, E. G.: Partial Klüver-Bucy syndrome produced by cortical disconnection. *Brain Res.* **16**, 281–284 (1969).
- HOREL, J. A. and MISANTONE, L. J.: The Klüver-Bucy syndrome produced by partial isolation of the temporal lobe. *Exp. Neurol.* **42**, 101–112 (1974).
- HUMPHREY, T.: The telencephalon of the bat. I. The non-cortical nuclear masses and certain pertinent fiber connections. *J. Comp. Neurol.* **65**, 603–711 (1936).
- JOHNSTON, J. B.: Further contributions to the study of the evolution of the forebrain. *J. Comp. Neurol.* **35**, 337–481 (1923).
- JONES, B. and MISHKIN, M.: Limbic lesions and the problem of stimulus-reinforcement associations. *Exp. Neurol.* **36**, 362–377 (1972).
- KEVETTER, G. A. and WINANS, S. S.: Connections of the corticomedial amygdala in the golden hamster. II. Efferents of the "olfactory amygdala". *J. Comp. Neurol.* **197**, 99–111 (1981).
- KRETTEK, J. E. and PRICE, J. L.: A description of the amygdaloid complex in the rat and cat with observations on intraamygdaloid axonal connections. *J. Comp. Neurol.* **178**, 255–280 (1978).
- LARSSON, K.: Features of the neuroendocrine regulation of masculine sexual behavior. In: BEYER, C. (ed.), *Endocrine Control of Sexual Behavior*, pp. 77–163. Raven Press, New York (1979).
- LEHMAN, M. N. and WINANS, S. S.: Vomeronasal and olfactory pathways to the amygdala controlling male hamster sexual behavior: autoradiographic and behavioral analyses. *Brain Res.* **240**, 27–41 (1982).
- LOO, Y. T.: The forebrain of the opossum, *Didelphis virginiana*. II. Histology. *J. Comp. Neurol.* **52**, 1–148 (1931).
- MACCHI, G.: The ontogenetic development of the olfactory telencephalon in man. *J. Comp. Neurol.* **95**, 245–305 (1951).
- MALSBURY, C. W. and PFAFF, D. S.: Neural and hormonal determinants of mating behavior in adult male rats. A review. In: DICARA, L. V. (ed.), *Limbic and Autonomic Nervous Systems Research*, pp. 85–136. Plenum Press, New York–London (1974).
- MISHKIN, M. and AGGLETON, J.: Multiple functional contributions of the amygdala in the monkey. In: BEN-ARI, Y. (ed.), *The Amygdaloid Complex*, pp. 409–420. INSERM Symposium No. 20. Elsevier/North-Holland Biomedical Press, Amsterdam–New York–Oxford (1981).
- MYERS, R. E.: Role of prefrontal and anterior temporal cortex in social behavior and affect in monkeys. *Acta Neurobiol. exp. (Warszawa)* **32**, 567–579 (1972).
- NIEMER, T. W. and GOODFELLOW, E. F.: Neocortical influence on the amygdala. *Electroenceph. clin. Neurophysiol.* **21**, 429–436 (1966).
- OTTERSEN, O. P.: The afferent connections of the amygdala of the rat as studied with retrograde transport of horseradish peroxidase. In: BEN-ARI, Y. (ed.), *The Amygdaloid Complex*, pp. 91–104. INSERM Symposium No. 20. Elsevier/North-Holland Biomedical Press, Amsterdam–New York–Oxford (1981).
- OTTERSEN, O. P.: Connections of the amygdala of the rat. IV: Corticoamygdaloid and intraamygdaloid connections as studied with axonal transport of horseradish peroxidase. *J. Comp. Neurol.* **205**, 30–48 (1982).
- OTTERSEN, O. P. and BEN-ARI, Y.: Pontine and mesencephalic afferents to the central nucleus of the amygdala of the rat. *Neurosci. Lett.* **8**, 329–334 (1978).
- PRELEVIĆ, S., BURNHAM, W. and GLOOR, P.: A microelectrode study of amygdaloid afferents: temporal neocortical inputs. *Brain Res.* **105**, 437–457 (1976).
- PRIBRAM, K. H. and BAGSHAW, M.: Further analysis of the temporal lobe syndrome utilizing fronto-temporal ablations. *J. Comp. Neurol.* **99**, 347–375 (1953).
- PRICE, J. L. and AMARAL, D. G.: An autoradiographic study of the projections of the central nucleus of the monkey amygdala. *J. Neurosci.* **1**, 1242–1259 (1981).
- RAISMAN, G.: An experimental study of the projection of the amygdala to the accessory olfactory bulb and its relationship to the concept of a dual olfactory system. *Exp. Brain Res.* **14**, 395–408 (1972).
- RUSSCHEN, F. T.: Amygdalopetal projections in the cat. I. Cortical afferent connections. A study with retrograde and anterograde tracing techniques. *J. Comp. Neurol.* **206**, 159–179 (1982).
- SCALIA, F. and WINANS, S. S.: The differential projections of the olfactory bulb and accessory olfactory bulb in mammals. *J. Comp. Neurol.* **161**, 31–55 (1975).
- SCHWABER, J. S., KAPP, B. S., HIGGINS, G. A. and RAPP, P. R.: Amygdaloid and basal forebrain direct connections with the nucleus of the solitary tract and the dorsal motor nucleus. *J. Neurosci.* **2**, 1424–1438 (1982).
- SHAMMAH-LAGNADO, S. J. and NEGRAO, N.: Efferent connections of the olfactory bulb in the opossum (*Didelphis marsupialis aurita*): A Fink-Heimer study. *J. Comp. Neurol.* **201**, 51–63 (1981).
- STEPHAN, H.: Der Bulbus olfactorius accessorius bei Insektivoren und Primaten. *Acta anat.* **62**, 215–253 (1965).
- STEPHAN, H.: Allocortex. In: BARGMANN, W. (ed.), *Handbuch der mikroskopischen Anatomie des Menschen*, 4. Bd., 998 Seiten. Springer, Berlin–Göttingen–Heidelberg (1975).
- STEPHAN, H.: Evolutionary trends in limbic structures. *Neurosci. Biobehav. Rev.* **7**, 367–374 (1983).
- STEPHAN, H. and ANDY, O. J.: Quantitative comparisons of brain structures from insectivores to primates. *Amer. Zool.* **4**, 59–74 (1964).

- STEPHAN, H. and ANDY, O. J.: Quantitative comparison of the amygdala in insectivores and primates. *Acta anat.* **98**, 130—153 (1977).
- STEPHAN, H., BARON, G. and FRAHM, H. D.: Comparison of brain structure volumes in Insectivora and Primates. II. Accessory olfactory bulb (AOB). *J. Hirnforsch.* **23**, 575 to 591 (1982).
- STEPHAN, H., BARON, G. and FRAHM, H. D.: Comparative size of brains and brain components. In: STEKLIS, H. (ed.), *Comparative Primate Biology*, Vol. III, Neurosciences, in press. Liss, Inc., New York (1987).
- STEPHAN, H., FRAHM, H. and BARON, G.: New and revised data on volumes of brain structures in insectivores and primates. *Folia primatol.* **35**, 1—29 (1981).
- STEPHAN, H., FRAHM, H. D. and BARON, G.: Comparison of brain structure volumes in Insectivora and Primates. IV. Non-cortical visual structures. *J. Hirnforsch.* **25**, 385—403 (1984).
- TAKEUCHI, Y., MATSUSHIMA, S., MATSUSHIMA, R. and HOPKINS, D. A.: Direct amygdaloid projections to the dorsal motor nucleus of the vagus nerve: A light and electron microscopic study in the rat. *Brain Res.* **280**, 143—147 (1983).
- TEN DONKELAAR, H. J., LAMMERS, G. J. and GRIENAU, A. A. M.: Neurogenesis in the amygdaloid nuclear complex in a rodent (the Chinese hamster). *Brain Res.* **165**, 348—353 (1979).
- TURNER, B. H., MISHKIN, M. and KNAPP, M.: Organization of the amygdalopetal projections from modality-specific cortical association areas in the monkey. *J. Comp. Neurol.* **191**, 515—543 (1980).
- VAN HOESEN, G. W.: The differential distribution, diversity and sprouting of cortical projections to the amygdala in the rhesus monkey. In: BEN-ARI, Y. (ed.), *The Amygdaloid Complex*, pp. 77—90, INSERM Symposium No. 20, Elsevier/North-Holland Biomedical Press, Amsterdam—New York—Oxford (1981).
- VÖLSCH, M.: Zur vergleichenden Anatomie des Mandelkerns und seiner Nachbargebilde. II. Teil. *Arch. Mikr. Anatom. u. Entwicklungsgeschichte* **76**, 373—523 (1911).
- WHITLOCK, D. G. and NAUTA, W. J. H.: Subcortical projections from the temporal neocortex in *Macaca mulatta*. *J. Comp. Neurol.* **106**, 183—212 (1956).

Authors' addresses:

Dr. H. STEPHAN
Dr. H. D. FRAHM
Max-Planck-Institut für Hirnforschung
Deutschordenstr. 46
D 6000 Frankfurt/M. 71

Prof. Dr. G. BARON
Département des Sciences biologiques
Université de Montréal
Case postale 6128, Succursale "A"
Montréal, Québec H3C 3J7, Canada